

The following problems will help you to prepare for the midterm exam. They indicate the style (but not the amount!) of problems that will form the bulk of the midterm: 1. giving examples, 2. correcting important definitions, 3. stating and illustrating important results, 4. proving smaller results, 5. ASN, 6-10. free response questions, mostly similar in content, style, and difficulty to the homework. I've emphasized content here from Ch 1 & 2, since you'll be working on many Ch 3 problems in the next week. A key for these review problems will be posted and problems will be discussed at the review session on Sunday, April 23.

1. Give an example of each of the following, or explain why it is not possible.
 - a) A matrix that is similar to an identity matrix but not row equivalent to it.
 - b) A matrix that is row equivalent to an identity matrix but not similar to it.
 - c) a 3×7 matrix with nullity 4
 - d) a 2×5 matrix with rank 3
 - e) A basis for $\mathbb{C}^3$ that contains no standard basis vectors
 - f) A span in $\mathbb{C}^3$ that contains no standard basis vectors
 - g) A vector space with a pair of subspaces that are disjoint
 - h) Four different examples of elementary matrices
 - i) a pair of matrices that are matrix equivalent but not row equivalent
 - j) a pair of matrices that are row equivalent but not matrix equivalent
 - k) A non-identity matrix A such that $A^2 = A^{-1}$.
 - l) A linear transformation with domain $\mathbb{R}^3$ and codomain $\mathbb{C}^2$
 - m) a pair of similar matrices with different determinants
 - n) a pair of matrices of different dimensions but the same determinant
 - o) three real 2×2 matrices with integer entries whose inverses also have integer entries
 - p) A matrix with nullspace equal to its column space
 - q) A matrix with nullity equal to its rank
 - r) five different fields
 - s) a pair of 2×2 matrices that have the same determinant but are not similar
 - t) a vector space with exactly two subspaces
 - u) a vector space with exactly three subspaces
 - v) a vector space with exactly five subspaces
 - w) a matrix that is upper triangular but not diagonal
 - x) a matrix that is both symmetric and antisymmetric
 - y) a linear transformation that is neither 1-to-1 nor onto
 - z) a map between two vector spaces that is not a linear transformation

2. Explain what is wrong about the following definitions and correct them.

A **vector space** is a set V of vectors with the structure of a group under addition, a field $\mathbb{F}$ of scalars, and a multiplication of vectors by scalars that distributes over addition of vectors.

A **linear combination** of vectors is a sum of scalar multiples of the vectors.

The **span** of a set of vectors in a vector space is the set of all of their scalar multiples.

A **linear transformation** is a function between vector spaces over the same field that respects the operations of vector addition and vector multiplication: $L(v + w) = L(v) + L(w)$ and $L(vw) = L(v)L(w)$.

A **basis** of a vector space V is a set of vectors in V such that any vector v in V admits a representation as a linear combination of the vectors in the basis.

A matrix is in **row echelon form** if every non-zero row has its leading non-zero entry strictly to the right of the first non-zero entry in the previous row.

The **range** of a linear transformation $L : V \rightarrow W$ is the subset of W sent to the zero vector in W by L .

Two matrices A and B are **similar** (or **conjugate**) if there exist invertible matrices P and Q such that $A = PBQ$.

3. Complete the following:

- a) State the Rank-Nullity Theorem. Illustrate the theorem with an example.
- b) Give four examples of conditions that are equivalent to a square matrix being invertible.
- c) State Cramer's rule. Illustrate the rule with an example.

4. Prove each of the following:

- a) Using the vector space axioms, show that $0\mathbf{v} = \mathbf{0}$ for any vector $\mathbf{v}$
- b) The intersection of two subspaces is a subspace.
- c) Using the defining properties of the determinant, show that a matrix with linearly dependent columns has determinant 0.
- d) If a matrix A represents a one-to-one linear transformation, then A^T represents an onto linear transformation.

5. Always, sometimes, or never? If it is always or never true, give a proof. If it is sometimes true, give an example where it's true and where it's false.

- a) An invertible matrix has determinant 1
- b) A lower triangular matrix is in RREF
- c) An upper triangular matrix is invertible
- d) Three vectors in $\mathbb{R}^3$ are linearly independent
- e) The vectors in a basis for a vector space V span a subspace of V .
- f) The sum of two matrices has fewer entries than either of the two matrices.
- g) The product of two matrices has fewer entries than either of the two matrices.
- h) An invertible matrix has an LU decomposition.
- i) Two distinct subspaces have different dimensions.
- j) When v and w are linearly independent, so are $v + w$ and $v - w$.
- k) A matrix is left invertible but not right invertible.
- l) A matrix is row equivalent to a matrix that is in reduced row echelon form.

6. For these two pairs of matrices, decide whether or not they are row equivalent. Explain your thinking.

$$A_1 = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 3 & 1 \\ 0 & 0 & -1 \end{pmatrix} \quad \& \quad A_2 = \begin{pmatrix} 1 & 0 & 0 \\ 4 & 9 & 0 \\ 6 & -3 & 10 \end{pmatrix}, \quad B_1 = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 5 & 6 \\ 0 & 0 & 9 \end{pmatrix} \quad \& \quad B_2 = \begin{pmatrix} 1 & 7 & 2 \\ 1 & 10 & 2 \\ 1 & -5 & 2 \end{pmatrix}$$

7. Give an example of a matrix that is similar to $\begin{pmatrix} 4 & 0 \\ 0 & 3 \end{pmatrix}$ but not equal to it.

8. Give a basis for each of the four fundamental subspaces of the following matrix: $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$

9. Give two different bases for the vector space $\mathcal{P}_2$ of real polynomials of degree at most 2. For each basis, give the matrix for the linear transformation that computes the derivative map $\mathcal{P}_2 \rightarrow \mathcal{P}_2$. Using each matrix, find a basis for the nullspace of the map.

10. For any n , define the $n \times n$ checkerboard matrix C_n as follows: $c_{i,j} = 1$ if $i + j$ is even, and 0 otherwise. For example, here is C_5 :

$$\begin{pmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \end{pmatrix}$$

Find a formula for $\det C_n$.

TEN IMPORTANT TERMS

A **vector space** is a set V of vectors with the structure of a commutative group under addition, a field $\mathbb{F}$ of scalars, and a multiplication of vectors by scalars that distributes over addition of vectors.

A **linear combination** of vectors is a finite sum of scalar multiples of the vectors. For vectors $v_1, v_2, \dots, v_p$, a linear combination of is a sum of form $\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_p v_p = \sum_1^p \alpha_k v_k$ where the α_k are scalars. The **span** of a set of vectors in a vector space is the set of all of their linear combinations.

A **linear transformation** is a function between vector spaces over the same field that respects the operations of vector addition and scalar multiplication: $L(\alpha v + \beta w) = \alpha L(v) + \beta L(w)$.

A **basis** of a vector space V is a set of vectors in V such that any vector v in V admits a unique representation as a linear combination of the vectors in the basis. The cardinality of a basis is the **dimension** of a vector space. (*Note: it is a theorem that dimension is well-defined.*)

A matrix is in **row echelon form** if all zero rows (if any) are below all non-zero rows, and if every non-zero row has its leading non-zero entry strictly to the right of the first non-zero entry in the previous row; the leading non-zero entry in each non-zero row is called its **pivot**. A matrix is in **reduced row echelon form** if it also has a 1 for each of its pivots and all entries above each pivot are 0.

The **kernel** of a linear transformation $L : V \rightarrow W$ is the subset of V sent to the zero vector in W by L . The **range** of L is the subset of W in the image of L . (*It is a result that the kernel and range are subspaces of V and W , respectively; they are also called the **null space** and **column space**).*

Two matrices A and B are **similar** (or **conjugate**) if there exists an invertible matrix Q such that $A = Q^{-1}BQ$.

For the square matrices $M_{n \times n}$ over a field $\mathbb{F}$, a **determinant** is a function $\det : M_{n \times n} \rightarrow \mathbb{F}$ such that $\det$ is multilinear and alternating and $\det(I) = 1$, where I is the $n \times n$ identity matrix. (*Note: it is a theorem that a determinant exists and is unique.*)

TEN IMPORTANT RESULTS: CHAPTERS 1-3

Any finite generating system for a vector space contains a basis. Every basis of a vector space has the same cardinality, yielding a well-defined dimension. (Prop. 1.2.8; Prop. 2.3.3)

Among the set of matrices that are row equivalent to a given matrix, there exists a unique matrix in reduced row echelon form. (Exercise 2.3.9; see handout.)

The solution set of a linear system $Ax = b$ is a translate of the solution set of $Ax = 0$ by a specific solution. (Thm. 2.6.1)

The Invertible Matrix Theorem. Let A be a square $n \times n$ matrix with entries in $\mathbb{F}$, and let $T : V \rightarrow V$ be the linear transformation $T(x) = Ax$. The following statements are equivalent.

- (1) A is invertible.
- (2) T is invertible.
- (3) A is row equivalent to I_n .
- (4) A has n pivots. (Prop. 2.3.6)
- (5a) $Ax = b$ has a unique solution for all b in $\mathbb{F}^n$. (Thm. 1.6.8)
- (5b) T is a bijection.
- (5c) The columns of A form a basis for $\mathbb{F}^n$. (Cor. 1.6.9)
- (6a) $Ax = 0$ has only the trivial solution. ($\text{Nul } A = \{0\}$)
- (6b) T is one-to-one.
- (6c) The columns of A are linearly independent.
- (7a) $Ax = b$ is consistent for all b in $\mathbb{F}^n$.
- (7b) T is onto.
- (7c) The columns of A span $\mathbb{F}^n$. ($\text{Col } A = \mathbb{F}^n$)
- (8) $\det(A) \neq 0$.
- (9) $0 \in \mathbb{F}$ is not an eigenvalue of A . (*not until Chapter 4*)

Rank Theorem (Thm. 2.7.1). For a matrix A , $\text{rank}(A) = \text{rank}(A^T)$.

Rank-Nullity Theorem (Thm. 2.7.2): For an $m \times n$ matrix A , $\text{rank}(A) + \dim(\text{Nul}(A)) = n$.

For any vector space of matrices, a determinant function exists and is unique.

Cofactor expansion of determinant (Thm 3.5.1) For an $n \times n$ matrix A , let $A_{j,k}$ denote the $(n-1) \times (n-1)$ matrix obtained from A by crossing out row number j and column number k . For any fixed j , we have

$$\det A = \sum_{k=1}^n a_{j,k}(-1)^{j+k} \det A_{j,k}.$$

Cramer's rule: For an invertible matrix A , the k th entry x_k of the solution of the equation $Ax = b$ is given by the formula

$$x_k = \frac{\det B_k}{\det A},$$

where the matrix B_k is obtained from A by replacing column number k of A by the vector b .

For $n \times n$ matrices A and B , $\det(AB) = \det(A)\det(B)$ and $\det(A^{-1}) = \det(A)^{-1}$ (Thm. 3.3.5)

Chapters 1 and 2:

vector
scalar
group
field
matrix
vector space
linear transformation
subspace
transpose
linear combination
trivial (linear combination, solution, subspace)
linear independence (and dependence)
span (generate)
spanning system
basis
standard basis
coordinates (in a basis)
dimension
isomorphism
rank
column space (range)
null space (kernel)
row space
one-to-one (injective)
onto (surjective)
isomorphism
conjugation, change of basis, similarity
homogeneous and inhomogeneous systems
vector equation
matrix equation
inconsistent and consistent systems
unique solution
trivial solution
general solution
coefficient matrix

augmented matrix
row operations
row equivalent
row reduction
row echelon form
reduced row echelon form (RREF)
pivot
free variable
(parametric) vector form
fundamental subspaces of a matrix
matrix equivalent

types of matrices/linear transformations:

zero
identity
elementary
projection
scaling
shear
rotation
permutation
diagonal
upper triangular
lower triangular
invertible (nonsingular)
non-invertible (singular)
symmetric (equal to transpose)

Chapter 3:

determinant
parallelepiped
submatrix
minor of order k
cofactor expansion
multilinear
antisymmetric or alternating
normalized